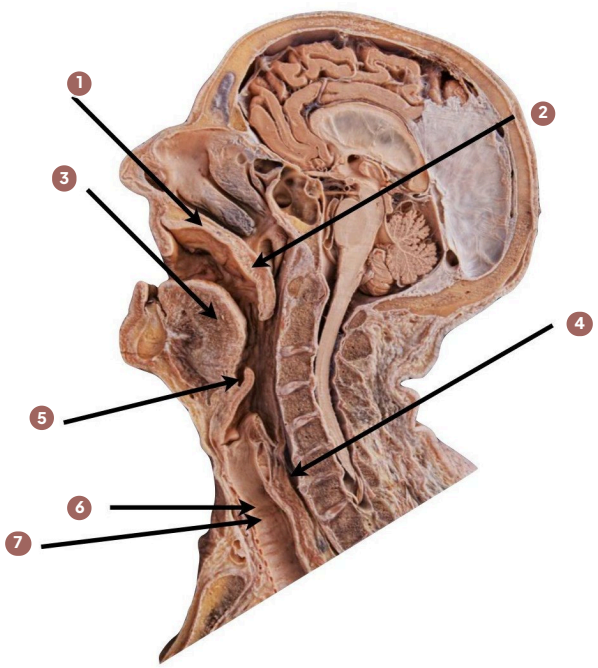
Respiratory System, Part 1

Competencies 97 to 99: airway map, nose and pharynx, larynx. Cadaver beside model. Fill in as we go.

Name _____

A. Nose, pharynx, larynx: the start of the airway (cadaver) COMPETENCY 97, 98



Cadaver, midsagittal head and neck

1. Hard palate
2. Soft palate
3. Tongue
4. Esophagus
5. Epiglottis
6. Vestibular folds (false vocal folds)
7. Vocal folds (true vocal cords)

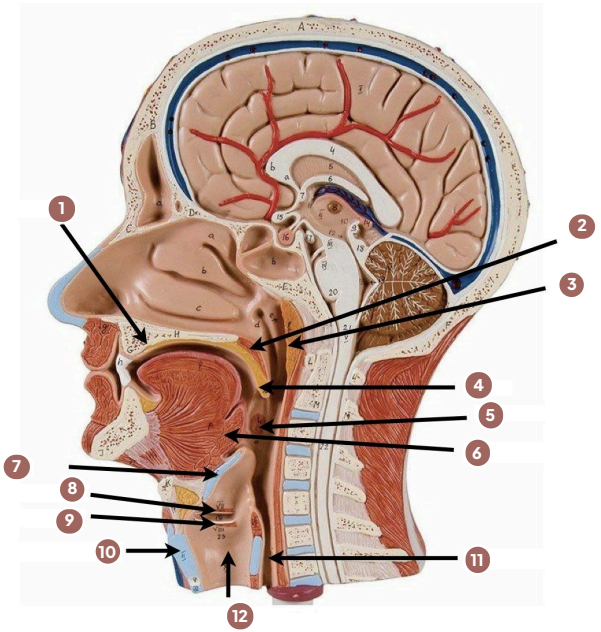
FILL IN

Air only region: nasopharynx

Air and food: oropharynx, laryngopharynx

Lower tract begins at the larynx

B. Palate, tonsils and the three regions of the pharynx (model) COMPETENCY 98, 113



Model, midsagittal head: palate, tonsils, larynx

1. Hard palate
2. Soft palate
3. Pharyngeal tonsil
4. Uvula
5. Palatine tonsil
6. Lingual tonsil
7. Epiglottis
8. Vestibular folds (false vocal folds)
9. Vocal folds (true vocal cords)
10. Thyroid cartilage
11. Esophagus
12. Larynx

PATHWAY OF ONE BREATH, IN ORDER

1. nares
2. nasal cavity
3. choanae
4. nasopharynx
5. oropharynx
6. laryngopharynx
7. larynx (glottis)
8. trachea

Extra notes and drawings

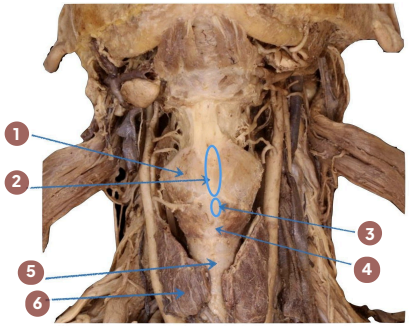
Extra notes and drawings

Respiratory System, Part 1

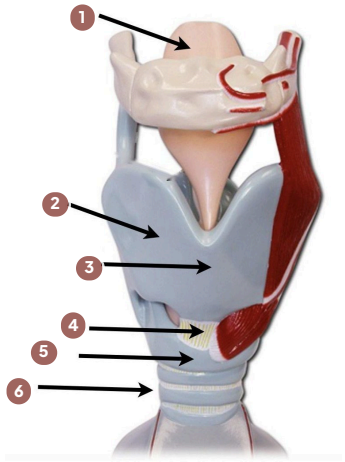
Competencies 97 to 99: airway map, nose and pharynx, larynx. Cadaver beside model. Fill in as we go.

Name _____

C. Larynx, anterior: cadaver beside model COMPETENCY 99



Cadaver, larynx and trachea, anterior



Model, larynx, anterior

	CADAVER	MODEL
1	Thyroid cartilage	Epiglottis
2	Laryngeal prominence	Thyroid cartilage
3	Cricothyroid membrane	Laryngeal prominence
4	Cricoid cartilage	Cricothyroid membrane
5	Trachea	Cricoid cartilage
6	Thyroid gland	Trachea

FILL IN

Largest cartilage, the shield: thyroid

Its front ridge: laryngeal prominence

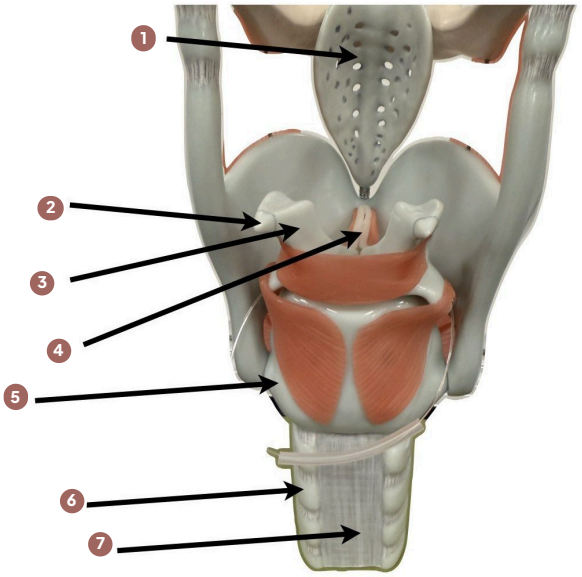
Only complete ring: cricoid

Emergency airway landmark: cricothyroid ligament

Single: thyroid, cricoid, epiglottis

Paired: arytenoid, corniculate, cuneiform

D. Larynx, posterior (model) COMPETENCY 99



Model, larynx, posterior

1. Epiglottis
2. Corniculate cartilage
3. Arytenoid cartilage
4. Vocal folds
5. Cricoid cartilage
6. Tracheal cartilages (C-rings)
7. Trachealis muscle

FILL IN

Elastic cartilage, not hyaline: epiglottis

Folds that make sound: vocal folds (true cords)

Folds that do not: vestibular folds (false cords)

Glottis = vocal folds plus the opening

Cartilage that pivots the folds: arytenoid

Open side of the C rings faces back, toward the esophagus

Extra notes and drawings

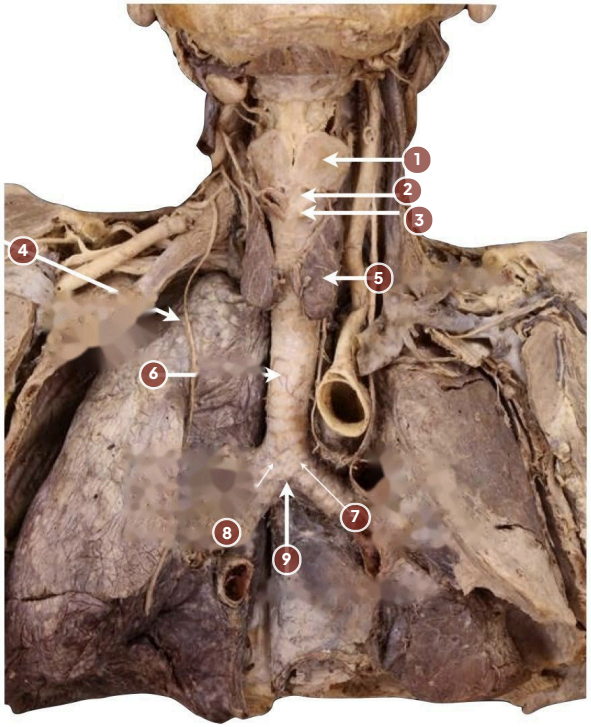
Extra notes and drawings

Respiratory System, Part 2

Competencies 100 and 101: trachea, bronchial tree, lungs and pleurae. Cadaver beside model. Fill in as we go.

Name _____

A. Trachea, carina and main bronchi (cadaver) COMPETENCY 100, 103



Cadaver, neck and thorax, anterior

1. Thyroid cartilage
2. Cricothyroid membrane
3. Cricoid cartilage
4. Phrenic nerve
5. Thyroid gland
6. Trachea
7. Left primary bronchus
8. Right primary bronchus
9. Carina

FILL IN

Carina sits at **the sternal angle, T4 to T5**

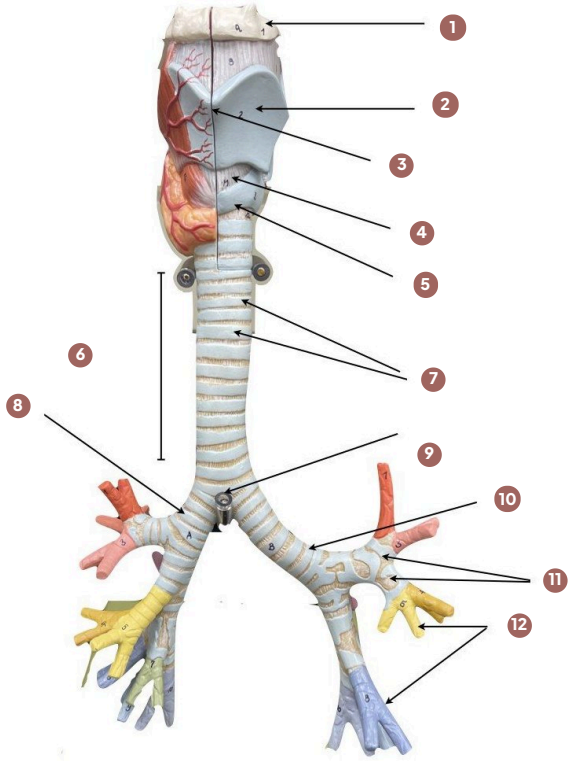
Right main bronchus is **wider, shorter, more vertical**

so an inhaled object lands **in the right lung**

Nerve to the diaphragm: **phrenic, C3 to C5**

Extra notes and drawings

B. The bronchial tree, trunk to twigs (model) COMPETENCY 100



Model, larynx, trachea and bronchial tree

1. Hyoid bone
2. Thyroid cartilage
3. Laryngeal prominence
4. Cricothyroid ligament
5. Cricoid cartilage
6. Trachea
7. Tracheal cartilages (C-rings)
8. Right primary bronchus
9. Carina
10. Left primary bronchus
11. Secondary (lobar) bronchi
12. Tertiary (segmental) bronchi

DOWN THE TREE, IN ORDER

1. trachea
2. main (primary) bronchi
3. lobar (secondary) bronchi
4. segmental (tertiary) bronchi
5. bronchioles
6. terminal bronchioles

FILL IN

Cartilage: rings, then **plates, then none**

Smooth muscle **increases**

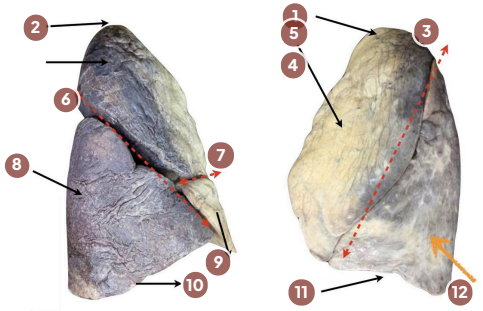
Extra notes and drawings

Respiratory System, Part 2

Competencies 100 and 101: trachea, bronchial tree, lungs and pleurae. Cadaver beside model. Fill in as we go.

Name _____

C. Lobes and fissures (cadaver) COMPETENCY 101



Cadaver, right and left lungs, anterior

1. Apex
2. Apex
3. Oblique fissure
4. Superior lobe
5. Superior lobe
6. Oblique fissure
7. Horizontal fissure
8. Inferior lobe
9. Middle lobe
10. Base
11. Base
12. Inferior lobe

FILL IN

Right lung lobes: superior, middle, inferior

Right fissures: horizontal and oblique

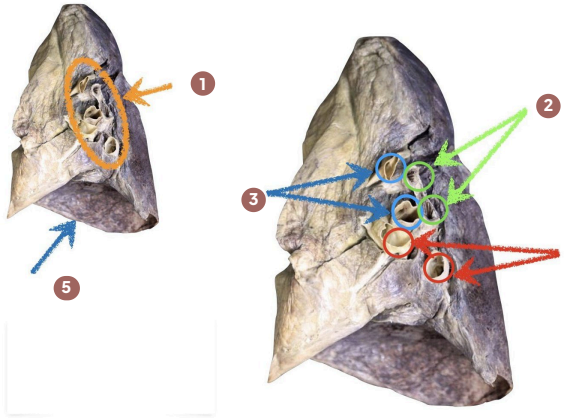
Left lung lobes: superior, inferior

Left fissure: oblique only

Left lung only: cardiac notch, lingula

Extra notes and drawings

D. Hilum and pleurae (cadaver, and the two-layer rule) COMPETENCY 101



Cadaver, right lung, medial view

ary arteries

ary veins

gmatic surface

layer

layer

cavity

led)

4. Pleural cavity
5. Parietal pleura
6. Visceral pleura

es through the hilum

arteries carry oxygen-poor blood in

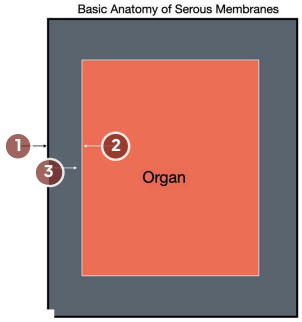
veins carry oxygen-rich blood out

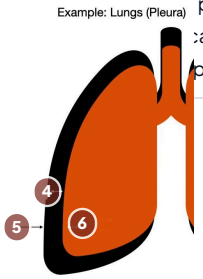
Parietal pleura lines the chest wall

pleura covers the lung

cavity holds a film of fluid

pocket: costodiaphragmatic recess





Serous membrane: parietal layer, visceral layer, cavity between

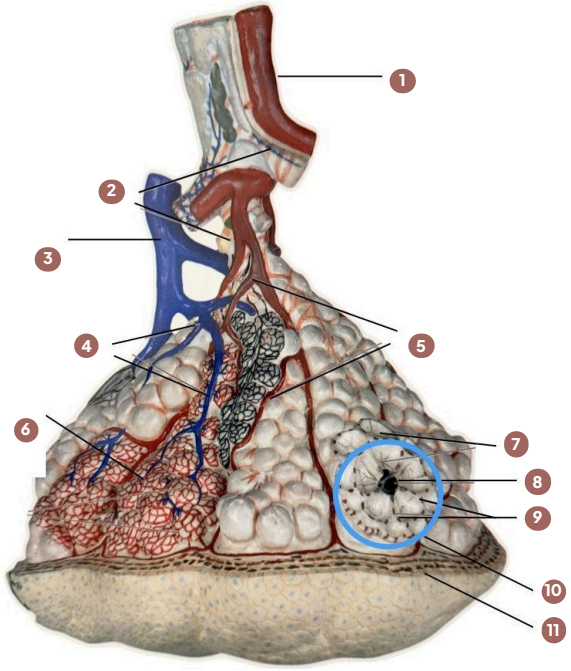
Extra notes and drawings

Respiratory System, Part 3

Competencies 102 to 104, with 97, 100 and 103: respiratory zone, histology, thoracic wall and diaphragm, disorders.

Name _____

A. Respiratory zone and the alveolus (model) COMPETENCY 97, 102



Model, lung lobule

1. Branches of pulmonary vein
2. Bronchioles
3. Branches of pulmonary artery
4. Pulmonary arterioles
5. Pulmonary venules
6. Capillaries surrounding alveoli
7. Alveolus
8. Opening of the alveolar duct
9. Pulmonary capillaries
10. Alveolar duct
11. Visceral pleura

FILL IN

Conducting zone ends at the: terminal bronchiole

Respiratory zone begins at the: respiratory bronchiole

Type I cell: thin and flat, gas crosses

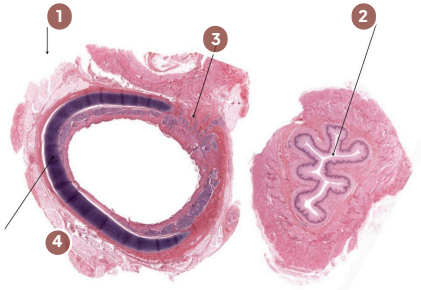
Type II cell: cuboidal, makes surfactant

Macrophage (dust cell): cleans up

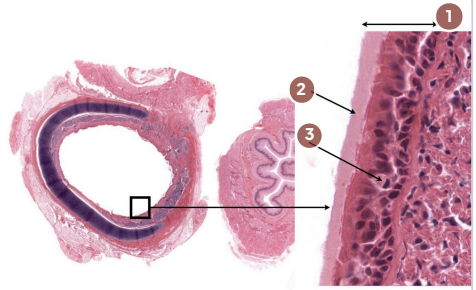
Membrane, air to blood: type I cell, fused basement membranes, endothelium

Extra notes and drawings

B. Histology: trachea and respiratory epithelium COMPETENCY 100, 102



Slide, trachea and esophagus, cross section



Slide, respiratory epithelium

1. Trachea
2. Esophagus
3. Trachealis muscle
4. Tracheal cartilage (hyaline cartilage)

1. Pseudostratified ciliated columnar epithelium
2. Cilia
3. Goblet cell

FILL IN

Open side of each C ring faces the: esophagus

Trachea epithelium: pseudostratified ciliated columnar

Goblet cells make: mucus

Cilia sweep it toward the: pharynx

Alveolar wall epithelium: simple squamous

Extra notes and drawings

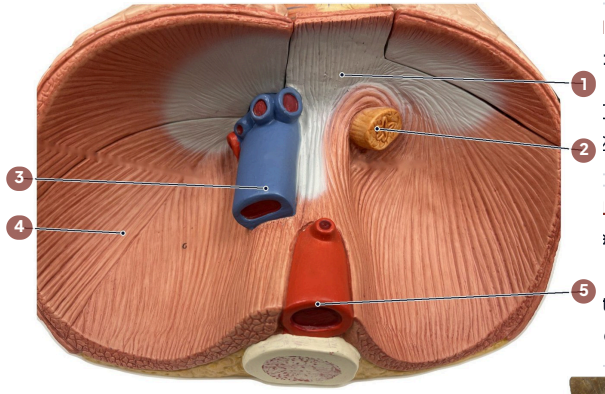
Respiratory System, Part 3

Competencies 102 to 104, with 97, 100 and 103: respiratory zone, histology, thoracic wall and diaphragm, disorders.

Name _____

C. Thoracic wall and diaphragm: the three openings

COMPETENCY 103



Model, diaphragm from above

1. Central tendon
2. Esophagus
3. Inferior vena cava
4. Diaphragm
5. Aorta

ROUGH THE DIAPHRAGM, TOP TO BOTTOM

: inferior vena cava, caval opening

): esophagus, esophageal hiatus, with the vagus nerves

2: aorta, aortic hiatus, with the thoracic duct

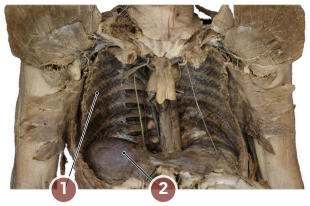
L IN

ain muscle of breathing: diaphragm

nerve and levels: phrenic nerve, C3, C4, C5

ternal intercostal fibers run down and forward

ernal intercostals work in forced expiration



Cadaver, thorax opened: the diaphragm below the lungs

1. Innermost intercostals
2. Diaphragm

Extra notes and drawings

D. Disorders: which structure does each one affect?

COMPETENCY 104

DISORDER: STRUCTURE, WHAT CHANGES

Asthma: bronchi and bronchioles narrow, no cartilage to hold them open

Emphysema: alveolar walls destroyed, permanent

Pneumonia: alveoli fill with fluid

Pneumothorax: air in the pleural cavity, lung collapses

Pleural effusion: fluid in the pleural cavity

COPD: emphysema plus chronic bronchitis

Infant RDS: type II cells, not enough surfactant

Lung cancer: lung tissue, tumor from airway epithelium

DRAW: ONE LUNG IN ITS TWO PLEURAL LAYERS, THEN SHOW A PNEUMOTHORAX AND AN EFFUSION ON IT

Extra notes and drawings